

KaarigarAI — Feature Decision Note

SIH26090 · AI-DRIVEN MARKET LINKAGE & SMART CATALOGING FOR MARGINALIZED ARTISANS

1. The Problem Statement

Government schemes fund artisans and weavers to set up small handicraft units, then give them a place to sell through periodic physical events — Shilp Samagam, Surajkund Mela, Dilli Haat. Those events produce a short burst of sales and then nothing. For the rest of the year these artisans have effectively no way to sell.

The reason is not that digital marketplaces refuse them. Amazon Karigar, GeM and ONDC all accept artisan sellers. The reason is that those platforms demand three things the artisan cannot produce: a clean product photograph, a well-written description in the right language, and a competitive price. Photography, copywriting and pricing are three separate professional skills, and a village artisan is unlikely to have any of them. Low literacy and language barriers make even the act of filling in a listing form impossible.

So the barrier is one of production and translation, not access. The problem statement asks for a mobile app that acts as a *virtual business manager* — one that substitutes for the photographer, the translator, the copywriter and the pricing analyst at once, and then connects the artisan onward to B2B buyers and government e-marketplaces. It mandates three features: an AI image enhancer, a multilingual voice-to-listing cataloger, and a dynamic pricing assistant. Everything must work for someone who cannot read or type.

2. What This Note Decides

The three mandated features are already designed. This note decides what goes *around* them. The existing plan is strong on cataloging and pricing but thin on the second half of the problem statement — actually putting the finished listing in front of a buyer. That gap is what most of the additions below are aimed at.

Sixteen candidate features were considered. Eight are recommended for build, three as small additions folded into existing screens, and five for the roadmap rather than the prototype. The numbers below are a triage list for this note only — they are not the feature numbers used in the proposal document. Each item carries two labels:

Integration — Simple / Intermediate / Hard. How much new engineering the feature needs on top of what already exists. *Simple* means it reuses components already in the build.

Impact — Low / Medium / High. How much the feature strengthens the submission, judged on both real-world usefulness and how well it demonstrates on stage.

BUILD THESE — eight features, roughly three days of work

Every one of these reuses something already in the plan. Together they close the market-linkage gap.

1. Voice-First First Run

Integration: Simple

Impact: High

- Registration without reading: language chosen by tapping a tile that speaks its own name aloud, phone number typed on a large numeric keypad, the one-time code read out instead of displayed.
- Nothing else is asked at first run. Craft type, region and bank details are collected later, only when a listing or a payment actually needs them.
- *Why build it: the whole design claims to work for someone who cannot read — and then the very first screen, before any AI can help, is the hardest literacy barrier in the product. This was the largest hole in the plan. Digits are the one symbol set most low-literacy users do reliably recognise, which is what makes a phone-number-only first run possible.*

NEEDS: three onboarding screens; pre-recorded audio of each language name; the text-to-speech component, which is shared infrastructure built before any feature that depends on it; permission to read the one-time password from SMS; an unverified account state allowing capture but not publication, so a user with no signal is never stranded on the first screen.

2. Offer Inbox and Buyer Catalog — "Bid-for-Craft"

Integration: Intermediate

Impact: High

- A buyer browses a plain public catalog by craft category and region, opens a listing, and sends one price offer. The artisan taps Accept or Reject. It is an inbox, not a live auction.
- Offers below her asking price, less a small tolerance, are auto-declined before she sees them — so she is protected from lowballing without judging anything. The fair-price floor is a separate mechanism: advice to her while pricing, not a gate on buyers.
- *Why build it: the problem statement explicitly asks the app to "connect directly with larger B2B buyers." At present the plan only suggests marketplaces. This is the weakest row in the requirement mapping, and this fixes it.*

Rating correction. This was first rated *Simple*, which quietly borrowed from a dependency: a buyer needs somewhere to send an offer from, so the public catalog and listing pages are part of this feature, not a separate one. Rated *Intermediate* and treated as a single shippable unit. Build the catalog first, the inbox second — neither is useful alone.

NEEDS: an offers table in the single system of record; one validation rule in the backend comparing an offer against the asking price; a push notification on arrival, with the app refreshing from the API rather than reading a second live datastore; one new app screen with large numbers and two buttons; a category-browsable public catalog and one permanent listing page per item, served by the existing backend from one plain template; a hosting tier with a stable domain, because a printed stall QR must still resolve months later — a tunnel is a demonstration tool, never the published URL.

3. Maker Story

Integration: Simple

Impact: High

- A short voice-narrated clip attached to the listing — who made this, where, and how — auto-transcribed and translated into both languages, and playable by the buyer.
- Recorded once per artisan and reused across all her listings, rather than re-recorded for every item.
- *Why build it: for handicrafts the story is not decoration, it is the margin. Provenance is the only reason a hand-thrown diya outsells a factory-moulded one at several times the price, and a bare product photograph destroys exactly that. It reuses the voice pipeline unchanged, so it is close to free, and it demonstrates beautifully.*

NEEDS: the existing voice pipeline; an audio clip stored against the artisan rather than the listing; the same personal-information stripping used on descriptions; an audio player on the public listing page.

4. Voice Order & Offer Status

Integration: Simple

Impact: High

- The artisan speaks a question — "Mera order kahan hai?" — and hears a spoken answer back.
- It runs the existing voice pipeline in reverse: same speech recognition, same language model, output spoken instead of written.
- *Why build it: right now the app listens by voice but answers in text. An artisan who cannot read still cannot use the second half of it. This makes the low-literacy claim actually true rather than aspirational.*

NEEDS: the existing speech-to-text stage; a small intent matcher for four or five fixed questions (order status, offers, price, earnings); the existing language model to phrase the answer; one new component — a free offline text-to-speech voice for Hindi and one regional language.

5. Marketplace Export Bundle

Integration: Simple

Impact: High

- One tap turns a finished listing into a ready-to-upload package for GeM, ONDC, Amazon Karigar and ODOP channels.
- No AI is involved. It reformats data the app already holds into the shape those platforms ask for, with images resized to their requirements.
- *Why build it: the largest return for the least effort in this entire list. It converts "we suggest a marketplace" into "here is the upload file," which is a materially different claim in front of judges.*

NEEDS: a small hand-written mapping of your product categories to each platform's category names, ODOP included; a spreadsheet and data-file writer; an image resizer for platform-specific dimensions; a zip packager; a share or download button.

6. Seasonal, Mela and Scheme Alerts

Integration: Simple

Impact: Medium-High

- A notification arrives unprompted: "Diwali is in three weeks — terracotta diyas are in demand. List now."
- The same calendar also carries the real events the problem statement itself names — "Surajkund Mela registration has opened," "the ODOP listing window closes Friday." Same notification, richer data.
- *Why build it: it makes the app behave like a manager who tells you things, rather than a tool you must remember to open. Almost free, given the seasonal logic already exists.*

NEEDS: a festival, season, mela and scheme-window calendar mapped to craft categories; a scheduled job that checks the calendar daily; the push-notification service already in the stack; a voice version of the alert reusing the text-to-speech from feature 2.

7. WhatsApp Share Card

Integration: Simple

Impact: Medium-High

- Each finished listing becomes a single shareable image — product photo, title in both languages, price band, and a QR code — that the artisan forwards on WhatsApp themselves.
- It uses the phone's normal share button. There is nothing to connect, register or get approved.
- *Why build it: it delivers the WhatsApp story, which matters because artisans already live on WhatsApp, without any of the WhatsApp engineering risk described in feature 12.*

NEEDS: a card-image generator (a fixed template filled with the listing's photo, text and price); the phone's native share sheet; the QR generator from feature 6.

8. Stall QR Tag

Integration: Simple

Impact: Medium-High

- A printable QR poster for the artisan's stall at a fair. A visitor scans it and lands on the permanent digital listing.
- It turns a one-day fair visitor into a year-round contact who can reorder after the event ends.
- *Why build it: the problem statement's whole narrative is that fairs give a temporary boost and then leave nothing behind. This is the bridge between the two channels, and it is roughly an hour of work. It also makes a strong closing beat in the demo.*

NEEDS: a QR-code library; a public web address per listing (shared with feature 10); a printable poster layout the artisan or a field officer can print at any shop.

FOLD IN QUIETLY — three small additions, no section of their own

These are cheap enough to include but not substantial enough to present as features. They belong inside screens that already exist.

9. Micro-Loan Eligibility Nudge

Integration: Simple

Impact: Medium

- A message appears: "Your sales may now qualify you for a higher PM Vishwakarma loan tier."
- It is one additional rule inside the scheme-matching assistant already planned, not a new feature.
- *Why include it: it ties the app to a real government scheme and reinforces the "business manager" framing at essentially zero cost.*

NEEDS: a running total of sales per artisan; two or three threshold rules added to the existing scheme matcher; no new screen.

10. Trust Badge / Verified Artisan

Integration: Simple

Impact: Low-Medium

- A badge appears on the artisan's listings after a few completed sales or a scheme verification.
- There is no intelligence behind it — a flag switches on when a condition is met.
- *Why include it: it addresses a real trust gap for buyers dealing with unfamiliar sellers, and it costs nothing. But it should not be oversold; judges recognise a badge as cosmetic.*

NEEDS: one true/false field on the artisan record; a rule that sets it; a badge icon on the listing view.

11. Listing Quality Meter

Integration: Simple

Impact: Medium

- A score out of 100 with spoken nudges: "add a close-up of the weave," "mention the material."
- Pure rules against a checklist — no model, no training data.
- *Why include it: this is what makes the app feel like a coach rather than a form. It is inexpensive personality, and it improves the listings the artisan actually produces.*

NEEDS: a checklist scoring function (photo count, description length, material mentioned, category set, texture shot present); a simple progress dial on the listing screen; the text-to-speech from feature 2 for the nudges.

SKIP — WRITE AS ROADMAP — five features, described but not built

Naming these deliberately is worth points. It shows the scope was chosen rather than missed. Each should get a short paragraph in the proposal marked "designed for, not yet built."

12. Full Live Auction

Integration: Hard

Impact: Medium

- A countdown timer, several buyers competing at once, highest bid wins at close.
- It requires handling people bidding at the same instant and ranking bids live — precisely the class of thing that breaks during a stage demo.
- *Why skip: the gain over a simple offer inbox is small, while the risk is large. Not a sensible trade with four weeks remaining.*

WOULD NEED: live socket connections, protection against simultaneous bids, a server-side auction clock, live bid ranking, and a tested tie-breaking rule.

13. WhatsApp Inbound Listing

Integration: Hard

Impact: High

- The artisan messages a photo and a voice note to a WhatsApp number instead of opening the app at all.
- It sounds free, and the messaging tier is — but it needs business-account verification, message-template approval and a permanently reachable public web address. All plumbing, none of it AI.
- *Why skip: the highest chance in this list of consuming a full day and still failing on demo day. Feature 5 captures most of the value safely. The idea still earns credit as a stated roadmap item.*

WOULD NEED: a verified business messaging account, an approved message template, a public HTTPS webhook with stable uptime, media download and queueing, and a phone-number-to-artisan identity mapping.

14. Cooperative / Self-Help-Group Listing

Integration: Simple

Impact: Low (for a demo)

- Several artisans in one cluster list under a single shared storefront.
- This is genuinely how self-help groups operate, and existing government platforms already support group-level onboarding.
- *Why skip: it is easy to build but changes nothing that can be shown in a six-minute demo. One paragraph acknowledging it is the right amount of attention.*

WOULD NEED: a group identifier on each artisan and listing; a shared dashboard view; a rule for splitting proceeds.

15. Peer Benchmark — "What Are Others Selling"

Integration: Intermediate

Impact: Low-Medium

- An anonymised view of how similar products in the same category and region are priced and performing.
- It overlaps with the explainability layer already planned, which already tells the artisan why their suggested price is what it is.
- *Why skip: it answers the same question twice through different interfaces, and it needs believable peer data — meaning more synthetic-data work for a modest gain.*

WOULD NEED: an aggregation query over listings by category and region; a credible synthetic peer dataset; anonymisation rules; a new comparison screen.

16. Buyer Dispute and Expectation Flow

Integration: Simple

Impact: Low-Medium

- Structured recourse for a buyer who receives an item that does not match its listing.
- Roadmap only — but worth naming, because a two-sided market that protects only one side does not stay trusted for long.
- *Why skip: there is no realistic way to adjudicate disputes without payment handling, which is itself out of scope. Acknowledging it shows the buyer's trust was considered, not only the artisan's.*

WOULD NEED: a dispute record against an accepted offer; photo evidence upload; an escalation path to a human — realistically a field officer, not the app.

3. Edge Cases and Failure Modes

The features above describe what happens when everything goes right. What follows is the opposite — the situations where the app is most likely to break, mislead, or be misused. These matter more than usual here, because the target user is in a noisy village home on a cheap phone with intermittent signal, not in a testing lab. Cases marked **MUST HANDLE** are the ones that would cause visible failure in a demo or real harm to an artisan; the rest can be acknowledged in the document as known limitations.

A. Voice and language

Loud background — children, machinery, TV MUST HANDLE	Confidence check on the transcription; if it is low, ask for a re-record rather than generating a wrong listing silently.
Dialect or accent the model handles poorly	Never fail outright — accept the best transcription, then let the artisan confirm by hearing it read back.
Hindi and English mixed in one sentence	Extremely common in real speech. The pipeline must not treat it as an error; translate what it can and keep the rest.
A craft word transcribed wrongly — "bandhani" heard as something else MUST HANDLE	Fuzzy-match against the craft glossary and correct it before the description is written. This is the single most visible quality bug in the demo.
Recording is two seconds long, or four minutes long	Enforce a minimum and a maximum length, with a spoken prompt explaining why.
Item-specific detail spoken into the reusable maker story	Keep the story generic to the artisan and her craft; anything about a single item belongs in that item's description, or the story stops being reusable.
The artisan speaks a phone number or ID number aloud MUST HANDLE	Strip digit sequences that look like personal identifiers out of the public description. A listing should never leak an Aadhaar number.

B. Photographs

White or pale product on a white wall MUST HANDLE	Background removal will eat part of the product. Detect a low-contrast result, keep the original photo, and ask for a retake on a darker surface.
Several products in one frame	Offer a simple crop step rather than guessing which item is being listed.
A hand or a person holding the item	Very common in artisan photos, and often desirable for scale. Do not aggressively strip it.
Photo taken at night with no light	Enhancement has a ceiling. The quality meter should nudge for a daylight retake instead of shipping a dark listing.
Glare on brass, mirror-work or glazed pottery	Blown-out highlights cannot be recovered. Flag it and suggest shooting away from direct sun.
Photo is a screenshot or a photo of a printed picture	Acknowledge as a limitation; the pipeline will process it but the output will be poor.

C. Pricing

A craft category the model has never seen MUST HANDLE	Return a deliberately wide band and say plainly that confidence is low. Never show a confident single number for an unknown craft.
Raw material cost unknown, so the floor cannot be computed	Fall back to a category-level median floor rather than dropping the protection entirely.
A genuinely high-value piece — a hand-woven silk saree worth tens of thousands	Sits outside the reference data range. Cap the estimate, mark it as out-of-range, and invite a manual price.
The artisan wants to price below the fair floor MUST HANDLE	Warn clearly, but allow the override after confirmation. Hard-blocking a person's own livelihood decision is the wrong ethical call, and judges will ask about this.
Per-piece versus per-set confusion — "six diyas for two hundred"	Ask explicitly which one it is; otherwise every price is wrong by a factor of six.
Seasonal multiplier fires at the wrong time	Cap how much the multiplier can move a price, so a calendar bug cannot triple a listing.

D. Offers and buyers

No offers arrive for a week MUST HANDLE	A permanently empty inbox is how an app dies. Nudge with a reason — improve the photo, widen the price, list on a marketplace.
Buyer makes an offer and disappears	Offers expire automatically and are marked lapsed, so the artisan is never waiting on a dead lead.
Two buyers accept the same one-of-a-kind piece MUST HANDLE	Handicrafts are usually unique. Reserve the item the moment one offer is accepted and auto-decline the rest.
A buyer floods every listing with lowball offers	Rate-limit offers per phone number; the price floor already rejects the offers themselves.
Accepted offer never turns into payment MUST HANDLE	State the boundary honestly: the app records intent, not payment. Pretending otherwise is the fastest way to lose credibility in questioning.
Buyer and artisan cannot understand each other	Offers carry a fixed structure — price, quantity, date — rather than free text, so no translation is needed to act on one.

E. Connectivity and the device

Offline for several days and the queue keeps growing MUST HANDLE	Cap the queue, downscale images before queueing, sync oldest first, and never silently drop a listing the artisan made.
The same listing syncs twice on reconnect MUST HANDLE	Give every listing a local unique identifier so duplicates are discarded server-side. Classic offline-first bug.
App is killed mid-recording or mid-upload	Autosave a draft; resume where it stopped instead of starting over.
Entry-level phone with little storage or memory	Compress before saving, keep heavy AI work on the server, and keep the offline model footprint small.
One phone shared by a whole family or group	Simple profile switching, which is also the groundwork for the group-listing roadmap item.
Interface language differs from the spoken language	Treat them as two separate settings. Many artisans speak one language and recognise script in another, or none.

F. Trust and misuse

A middleman uses the app on the artisan's behalf and keeps the margin MUST HANDLE	The most serious real-world risk in this design. Tie the account and the stall QR to the artisan's own phone number, and show earnings in the artisan's own view.
Mass-produced factory goods listed as handmade	Out of scope to police properly, but worth a duplicate-photo check and an honest note in the limitations section.
The same photograph reused across many listings	An image fingerprint check catches this cheaply.
A false trust badge from self-dealt sales	Require distinct buyers before the badge appears, or the badge means nothing.
The AI writes something factually wrong about the craft MUST HANDLE	Read the description back aloud for confirmation before publishing. The artisan is the authority on their own craft, not the model.

G. First run and identity

The one-time password never arrives MUST HANDLE	Registration cannot dead-end at an SMS. Allow a retry, a voice-call code, and a resume-later state that keeps anything already captured.
A digit is mistyped in the phone number	Read the number back aloud digit by digit before sending the code. This is cheaper than any correction flow.
The wrong language is picked by accident	Language must be changeable from the first screen of the app afterwards, using the same speaking tiles — not buried in a settings menu the user cannot read.
The phone belongs to a family member MUST HANDLE	Common, and it is how earnings quietly end up attributed to the wrong person. Support more than one artisan profile per device from the start.

H. Demo day

No usable internet at the venue MUST HANDLE	Rehearse a fully offline path with local models and pre-seeded data. Assume the venue Wi-Fi fails.
A free-tier quota runs out mid-presentation MUST HANDLE	Cache the responses for the exact demo items so the flow completes regardless.
A judge asks to speak an untested language	Say which languages are tested and which are supported in principle. An honest boundary beats a live failure.
Speech recognition takes too long on stage	Keep a pre-recorded clip and a short loading animation, and narrate over the wait.
A judge asks what happens with no buyers on the platform MUST HANDLE	Have the answer ready: the marketplace export bundle means the app is useful with zero buyers of its own, because it feeds existing marketplaces.

4. The Shape of the Decision

Eight features built, three folded quietly into existing screens, five written up as roadmap. Total new build time is roughly three days, and every recommended feature is rated *Simple* to integrate except the offer inbox, which carries the public buyer catalog with it.

The through-line. Almost every feature recommended for build attacks the same half of the problem statement: **market linkage** — the half the original design under-serves. The offer inbox, the marketplace export bundle, the share card, the stall QR and the maker story are five expressions of one idea, which is getting the finished listing in front of an actual buyer and giving her a reason to pay more for it.

The two exceptions earn their place differently. **Voice-first first run** and **voice order status** exist because the low-literacy claim runs through every page of this proposal, and without them it breaks at the two points a user meets first and last.

The cataloging and pricing half of the problem statement is already well answered. This is what turns a very good listing tool into the virtual business manager the problem statement actually asked for.